

Project ID:

25-26J-174

1. Topic (12 words max)

Traffic and Accident Detection System

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max)
– references not included in word count.

In urban areas with increasing traffic density, traffic violations and accidents are a growing concern. Existing monitoring systems either lack real-time processing capabilities or fail to provide transparency in the enforcement and appeal process. This leads to delayed emergency response, inadequate evidence handling, and public dissatisfaction due to lack of accountability in issuing traffic fines.

With advancements in AI, image processing, and cloud computing, it is now feasible to implement a smart traffic monitoring system that not only detects violations but also identifies offenders and provides mechanisms for fair appeal handling. The challenge lies in integrating these modules into a unified, real-time and citizen-friendly system.

Our proposed solution aims to address these issues through an AI-driven camera surveillance system to detect specific violations (e.g., overtaking on single/double lines, red light jumping), vehicle and license plate identification using OCR, and a severity-based emergency response module. It also includes a citizen portal where users can view their fines, submit appeals with evidence, and track the appeal status with transparency. Notifications via SMS/email and a police checkpoint accountability module are included to ensure proper follow-up.

This multi-faceted approach provides not only technical innovation but also administrative transparency and real-world utility, making traffic enforcement smarter and fairer.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The system is divided into four major modules, all coordinated through a central server:

1. **Violation Monitoring System**

Cameras installed at accident-prone locations continuously monitor traffic. AI-based models detect specific violations, including overtaking on single or double lines and running red lights. Upon detecting a violation, the system triggers a response by sending the event to the server for further processing.

2. **License Plate & Vehicle Identification**

Upon receiving a violation trigger, this module uses OCR (Optical Character Recognition) to detect and read the vehicle's license plate. It then retrieves related vehicle data, such as the type, color, and registered owner's information. This information is logged and forwarded to the nearest police checkpoints for further action.

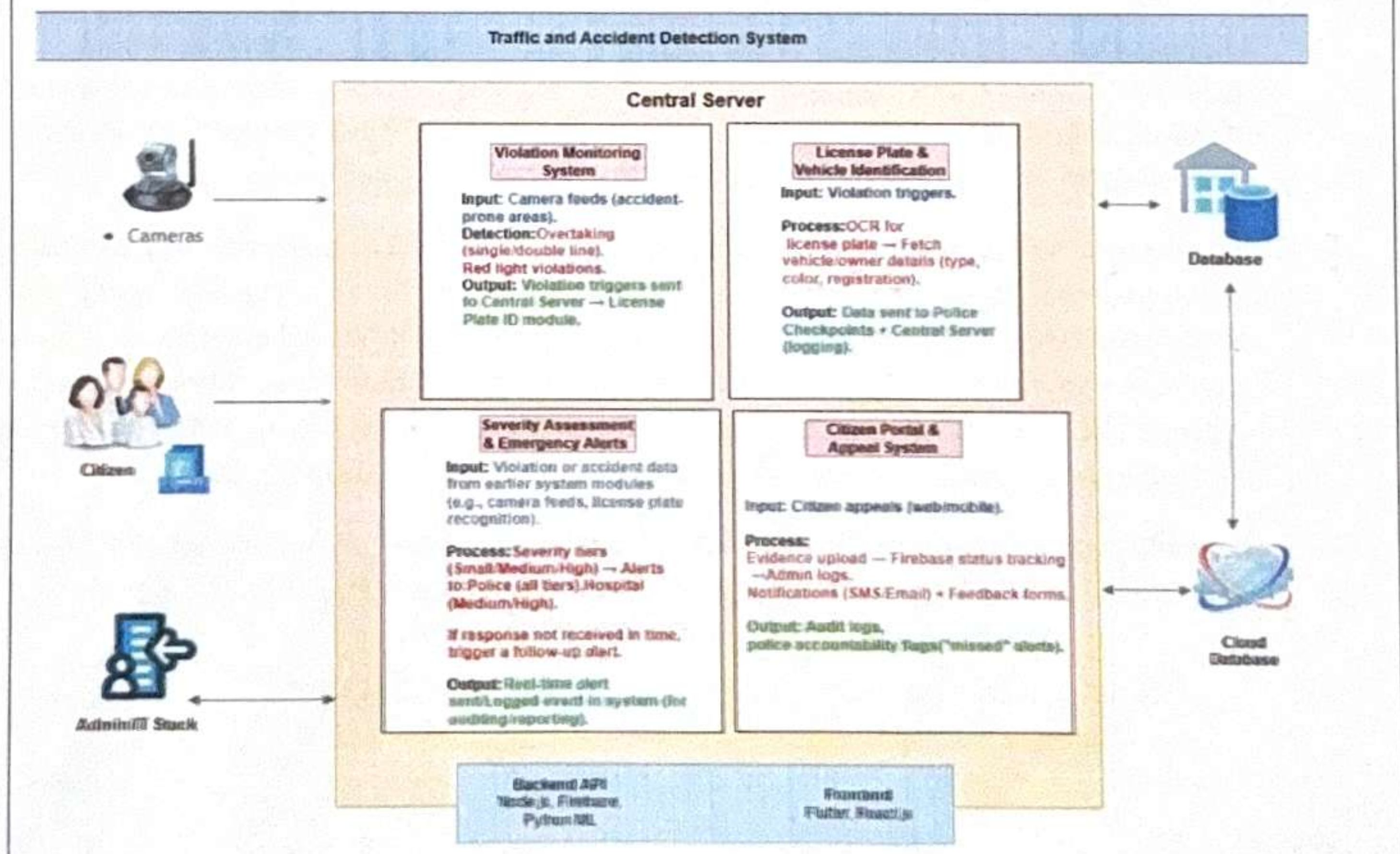
3. **Severity Assessment & Emergency Alerts**

Classifies incidents into small, medium, or high severity. Based on classification, it alerts police and hospitals. If no acknowledgment is received in a defined time, a follow-up alert is generated.

4. **Citizen Portal & Appeal System**

Allows users to view fines, submit appeals with evidence, and track status in real-time via Firebase. Admins can review appeals and respond. Users receive email/SMS notifications and can give feedback after resolution. Audit logs and officer accountability mechanisms are included.

The system uses Node.js, Python (ML) for backend, Firebase for real-time database and notifications, and Flutter/React.js for the frontend.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This solution requires interdisciplinary expertise in AI, computer vision, OCR, cloud infrastructure, and front-end/back-end development.

Technical Requirements:

- **Computer Vision** for violation detection (YOLO/OpenCV).
- **OCR** for number plate recognition (Tesseract).
- **Firebase** for real-time updates, Firestore for database.
- **Cloud functions/APIs** for communication between services.
- **Frontend frameworks:** Flutter (mobile) and React.js (web).
- **Alert systems** via Email/SMS (Twilio or Firebase messaging).

Data Requirements:

- Video datasets of real traffic for training detection models.
- Vehicle registry data for linking license numbers to owners.
- Citizen NIC and vehicle numbers for login/authentication.
- Police checkpoint geolocation data for alert routing.

This setup not only fulfills technical goals but also complies with real-world social governance, making the project highly impactful.

8. Objectives and Novelty

Main Objective Develop a real-time traffic violation and accident detection system integrated with emergency alerting and a transparent digital appeal mechanism.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Pushpamal K P N IT22561916	Real-time detection of critical traffic violations using AI	Integrate surveillance cameras, develop and train AI models to detect violations (e.g., overtaking, red light crossing)	Real-time AI-based detection of violation types (single-line overtaking, double-line overtaking, red-light jumping)
Wathsalavi K H S IT22305596	Identify violating vehicles and retrieve ownership details using OCR and vehicle databases	Detect license plates using OCR, retrieve vehicle type, color, and owner information from database, and forward data to police checkpoints	OCR-based license plate detection + Real-time vehicle data extraction + Automated police alerting
Rashmi B.G.R IT22052360	Assess severity of accidents or violations and alert emergency services	The system analyzes crash data to classify severity as Low, Medium, or High. It sends alerts to police for Low cases, and to both police and hospital for Medium or High cases. If no response is received within a set time, it resends the alert. All alerts and responses are logged for reference.	Severity-based emergency classification + Automated police & hospital alerting + Response time tracking with alert resending

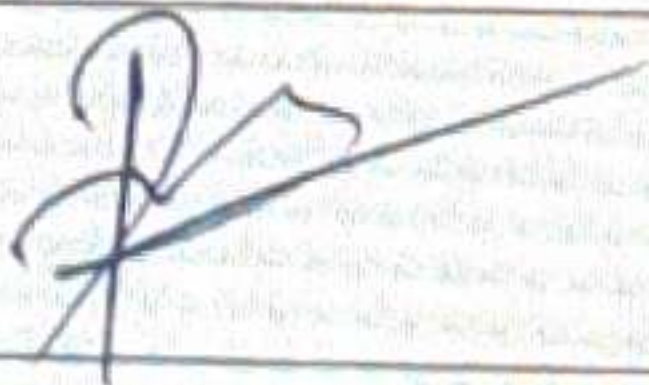
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Shehara I G D S IT22338334	Transparent appeal module with tracking	Appeal Status tracker, appeal approve or rejection explanation, email/SMS alerts indicating the results of the appeal, Admin dashboard logging, Feedback option for users (used to evaluate public satisfaction and improve the system in the future), Generate summary report of the appeal.	Real-time tracking + Admin logs + email/SMS alert auditing
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Pushpamal K P N It22561916	Implements AI-driven computer vision models for detecting real-time traffic violations such as overtaking and red-light offenses. This aligns with the Artificial Intelligence specialization through model training, detection logic, and intelligent decision-making.
Wathsalavi K H S IT22305596	Uses OCR and database integration to extract license plate numbers and map them to vehicle and owner details. This complies with Intelligent Systems and Embedded Technology specialization by combining image processing, data mapping, and real-time system response.
Rashmi B.G.R IT22052360	This component applies AI and automation techniques to classify accident severity and trigger real-time alerts to police and hospitals. It aligns with the Information Technology specialization by integrating real-time data processing, emergency response automation, and intelligent decision-making to enhance public safety systems.
Shehara I G D S IT22338334	Develops Firebase-backed frontend apps with transparency, reflecting full-stack/cloud computing focus.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Lecturer	Dinith	Primal	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

<i>This project is ok to proceed.</i>
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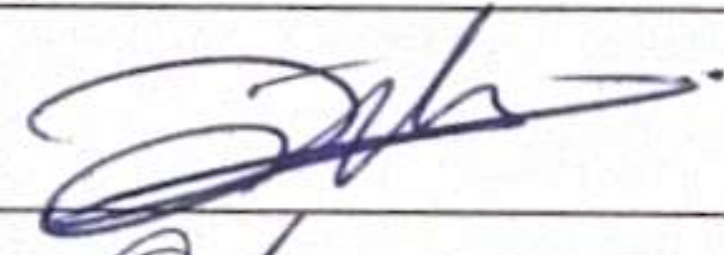
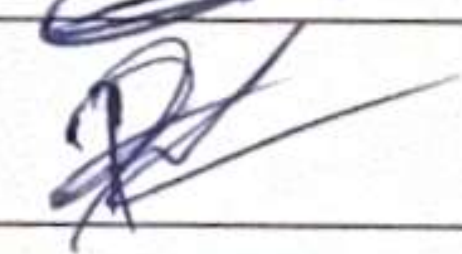
Acceptable: Mark/Select as necessary

Topic Assessment Accepted	✓
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

* You should strongly address the novelty of your project components.
 * You have to select specific junction and need justify the selection of the specific junction.
 * Collect the Data from police station.
 * Think about the real world implementation.
 * Think about the passengers inside the vehicle

Staff Member's Name	Signature
Mr. Deemantha Siriwardana	
Mr. Dinith Primal	

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.